Studying Interpretability in LLMs using Representational Decomposability

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Abstract

Post-hoc interpretability methods, such as probes and sparse autoencoders (SAEs), approximate the computation they explain. We study three architectural mechanisms that build interpretability into the model, and combine them in a single decoder-only language model: Backpack sense vectors, which make the logits an exact sum of token-level contributions; CoCoMix, which adds a prediction and feedback channel in the concept vocabulary of an SAE; and OrtSAE's orthogonality penalty, which we apply to every set of vectors the model uses as units. On a 1 Million parameter model trained on TinyStories dataset, we compare combinations of these ideas with a shared set of probes for readability, attribution faithfulness, intervention selectivity and training cost. From our experiments, exact additive decomposition yields faithful attributions and selective interventions that learned interfaces do not provide. The concept channel is writable, but its efficacy is bounded by the teacher's vocabulary. Orthogonality lowers the similarity between units without improving their readability. The loss in perplexity arises from mechanisms that reroute computation, not from added supervision, and it is recovered by widening the plain model.

1 Introduction

Recent advancements in Large Language Models (LLMs) have led to impressive performance across a wide range of tasks. However, the interpretability of these models remains a big challenge. Understanding how LLMs make predictions and how we can control their behavior is important forreliability and safety. Methods for understanding a language model fall into two families. Post-hoc methods, such as probes, sparse autoencoders (SAEs) and activation patching, explain a trained model with a second, fitted thing and can therefore only approximate the computation they describe. Built-in methods instead constrain the architecture or the objective so that the explanation is part of the forward pass, which allows it to be exact. This advantage is not free: structural constraints may reduce quality, and a structure that appears readable need not be readable in use. It remains unclear which built-in mechanisms deliver interpretability, at what cost, and whether they can be combined.

We study three mechanisms: Backpack language models [Hewitt et al., 2023], which decompose the output into an exact sum of context-free token senses; CoCoMix [Tack et al., 2025], which trains the model to predict and consume concepts from the dictionary of an SAE; and OrtSAE [Korznikov et al., 2025], which penalises similarity between dictionary entries. We integrate all three in one small decoder transformer model trained on TinyStories dataset. So we can inspect every component, and evaluate a combination set of configurations with a common set of evaluations along four dimensions: exactness of attribution, writability of the concept channel, geometric distinctness of units, and computational routing cost.

Our contributions are as follows. First, we describe a unified decoder in which concepts act only by re-weighting sense attention, which preserves the exact decomposition, and in which the orthogonality

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penalty applies to all kinds of units rather than only to SAE dictionaries. Second, we provide a controlled comparison of eight configurations, with controls that separate the cost of supervision from the cost of routing. Finally, we will report our findings.

2 Related Work and Background

Post-hoc interpretability explains a trained model through auxiliary objects fitted to its activations, most prominently SAEs [Bricken et al., 2023, Cunningham et al., 2023]. Built-in approaches instead modify the model so that a readable structure is part of its computation. We build on three such approaches and, throughout, let E denote the tied embedding matrix, so that the logits at position t are $E\,o_t$ for a final representation o_t. To our knowledge none of the three has been combined with the others or evaluated with the same probes.

2.1 Backpack Language Models

In a transformer [Vaswani et al., 2017] the output is a nonlinear function of the whole context, so any explanation of a prediction is an approximation. A Backpack [Hewitt et al., 2023] generalises the continuous bag of words of word2vec [Mikolov et al., 2013] to this setting. A transformer is retained for one purpose, computing how much each earlier token contributes, and all token-level content is moved into a lookup. Every token x has k *sense vectors* $S_1(x), \ldots, S_k(x)$ computed from the token alone, so a sense is a context-free property of the word. Sense vectors are learned without supervision or semantic priors. The context determines only non-negative weights $\alpha_{\ell ij}$, one attention pattern per sense, and the output is $o_i = \sum_{j \le i} \sum_{\ell} \alpha_{\ell ij} S_\ell(x_j)$. As no nonlinearity follows the sum,

$$E\,o_i = \sum_{j \le i} \sum_{\ell} \alpha_{\ell ij}\, E\, S_\ell(x_j). \tag{1}$$

This decomposition intrinsically provides three operations. A sense is *read* through the tokens it promotes, $E\,S_\ell(x)$, independently of context. A prediction is *attributed* because its logit is a sum of (source token, sense) terms of known size. A sense is *edited* by scaling, which changes only its own term and does so linearly.

2.2 CoCoMix

A model trained only for next-token prediction has no incentive to organise its hidden state into human-recognisable quantities. CoCoMix [Tack et al., 2025] obtains a vocabulary of such quantities from a teacher: an SAE is fitted to the teacher's hidden state and its latents are the *concepts*. At each token, an attribution score (the change in next-token loss caused by a concept's direction) selects the few most relevant concepts as labels. The student carries a linear head $z_t = M h_t$ at an intermediate layer, trained to predict these labels, and compresses its own top predictions into a single vector $\hat{c}_t = W\,\mathrm{TopK}(z_t) + b$ that is inserted into the sequence next to h_t for the remaining layers. The channel is therefore *read* (which concepts are active at a token) and *written* (adding to z_t changes what the later layers receive). Reading and writing take place in the space of the teacher's SAE, so the usefulness of the interface is bounded by the quality of that vocabulary. The original work uses a pretrained model as teacher; here the teacher is a plain model of the same size as the student, trained separately for a shorter schedule, which is a deliberately weak teacher.

2.3 Sparse Autoencoders and OrtSAE

An SAE writes a hidden state as $h \approx \sum_i c_i w_i$ with few non-zero coefficients c_i; we use the TopK variant [Gao et al., 2024]. The decomposition is readable only if each decoder row w_i has a distinct meaning, and two failures arise when rows are similar. In feature absorption [Chanin et al., 2024], a general row ("starts with S") is silent on words that have a more specific row of their own; in composition, one row is the sum of two more basic ones. OrtSAE [Korznikov et al., 2025] adds a penalty on the largest cosine similarity between each row and any other,

$$\Omega = \frac{1}{m} \sum_i \Big(\max_{j \ne i} \cos(w_i, w_j) \Big)^2, \tag{2}$$

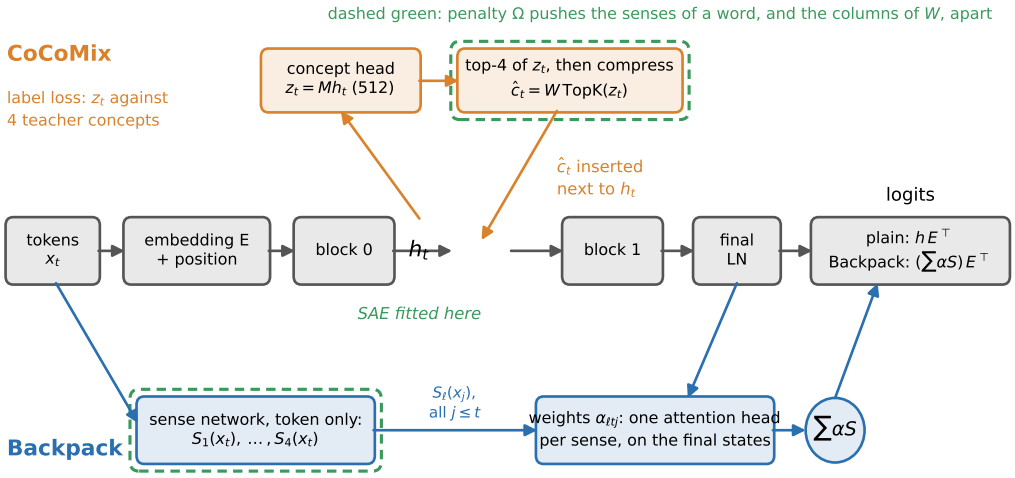


Figure 1: The three switches in one decoder. Blue: Backpack readout. Orange: CoCoMix concept head and feedback. Green: the orthogonality penalty, applied to the vectors it acts on.

which pushes rows apart. In contrast to the two mechanisms above, it does not change what the model computes: distinctness is a measurable property of a set of vectors that can be enforced and verified directly. Whether it makes the vectors easier to read is a separate, empirical question.

3 Method

All the models in this report share one pre-LayerNorm decoder: 2 blocks, 2 heads, d=128, context 256, tied embeddings and a 4096-token vocabulary (953,856 parameters in the plain model). Three switches modify it (Figure 1). The common site is h_t, the residual stream after block 0. It already holds the token and some context, one block remains to use what is written there, and every SAE is fitted at this site, so that the interfaces of different models are compared like for like.

We build on the mechanisms we talked about in Section 2, we integrate them as follows. Concepts reach the output only by re-weighting senses, so the decomposition of Eq. (1) remains exact. The orthogonality penalty is applied to every set of vectors the model uses as units (senses, concept directions and SAE rows), not only to an SAE. All models are then evaluated with the same probes of readability, attribution, intervention and cost.

Backpack readout (k=4 senses, +115,840 parameters). The sense network is a LayerNorm, a linear layer, a GELU and a linear layer from d to kd; it sees only the token embedding, never position or context. The weights come from k attention heads (width d/k) on the final hidden states, whose values are the sense vectors. There is no value or output projection, residual or LayerNorm after the sum, so the logits are exactly Eq. (1). We use four senses where the original uses sixteen, since this model is over a hundred times smaller; the aim is a small number of senses per word that can each be read.

Concept head (C=512, +131,712). $z_t = Mh_t$ is trained with cross-entropy against four teacher concepts per token (weight λ=0.1). Feeding it back is a separate switch: the four largest logits are compressed by W into one vector that is inserted after each h_t before block 1, and the output is read from the concept slots, as in the released code. Keeping the two switches apart yields a control that separates the cost of *supervision* (the label loss alone, *CE only*) from the cost of *routing* (feedback through the channel). In the combined model the concept slot is the query of the sense attention, so concepts decide which earlier words' senses are used but never what a sense says, and Eq. (1) stays exact.

Teacher and labels. The teacher is a plain model of the same size, trained separately for a shorter schedule. A TopK SAE (m=512, K=16) on its block-0 stream gives the concepts. At each token the

Table 1: The eight configurations and the factor each isolates.

Model	Senses	Concept head	Penalty Ω	Isolates
P	no	no	no	the reference, with only post-hoc SAEs
CC	no	label loss only	no	cost of supervision without routing
CM / CMo	no	fed back	no / on W	cost of routing through concepts; effect of Ω on concept directions
BP / BPo	4	no	no / on senses	the exact readout; effect of Ω on senses
BC / BCo	4	fed back	no / on both	concepts steering exact senses; both penalties

labels are the four latents with the largest $c_i\langle\nabla_h L, W_i^{\mathrm{dec}}\rangle$, i.e. those that most change the next-token loss (the released-code rule). The student's concepts are the teacher's latent indices.

Penalty. Ω of Eq. (2) with weight γ_o=0.1, applied to the vectors a model uses as units: the k senses of a random set of tokens, and the columns of W when concepts are fed back. After training, a TopK SAE and an OrtSAE (γ=0.25), both m=1024, K=16, are fitted to every model's h_t; they are the post-hoc interface that all models have in common.

Interfaces and model configurations. A model can offer up to three kinds of unit: senses (Backpack), concepts (concept head) and post-hoc SAE latents (every model). Table 1 lists the eight configurations compared. Each differs from a neighbour by one switch, so that a difference in an interface or in cost can be attributed to a single cause. P is the reference; CC and CM separate supervision from routing; BP and BC evaluate the readout alone and with concepts; the penalised variants (suffix o) test whether distinctness improves the units on which it acts.

4 Experimental Setup

Dataset and task. All models are trained from scratch on TinyStories [Eldan and Li, 2023], a corpus of short synthetic stories written by GPT-3.5 and GPT-4 in the vocabulary of a young child. The task is autoregressive language modelling: given the first t tokens of a text, predict token t+1, with the next-token cross-entropy (CE) as the training loss; where a switch of Table 1 is on, the concept-label loss and the penalty Ω are added (Section 3). Text is tokenised with a byte-level BPE vocabulary of 4096 tokens fitted on the first 100k stories, with one end-of-story token appended to every story. The stories are concatenated and cut into non-overlapping windows of 256 tokens, and every training step draws 128 windows at random. Stories are split into four disjoint sets. The *train* set (300k stories, 69.4M tokens) is the only data the language models are trained on; 30,000 steps correspond to 983M tokens, about 14 passes over it. The *sae* set (20k stories, 4.5M tokens) is used only to fit the post-hoc SAEs. The *dev* set (5k stories, 1.1M tokens) is used only to select the checkpoint with the lowest CE during training. The *test* set is the official validation split of TinyStories (21,990 stories, 4.9M tokens in 19,101 windows). It is out of sample: it shares no story with the other sets and is never used for training, for fitting SAEs, or for choosing a checkpoint. All reported numbers, namely test perplexity (the exponential of the mean test CE), the paired CE difference between models on identical test tokens, and every probe of Section 5, are computed on it. The concept labels of the CoCoMix models are computed by the teacher on the train windows (Section 3). The one use of test text in selecting an intervention is the choice of the best sense in the editing experiment, which is stated where it occurs (Section 5.2).

Models and training. We compare the eight configurations of Table 1. Each configuration is trained with four seeds. Figures report the mean over the four trained models with the standard error across them, and dots denote single models. Fifteen further variants (the remaining combinations of the three switches and two wider plain models used as capacity controls) are trained with one seed and reported only in the appendix (Appendix A).

Evaluation protocol. Four properties of a unit are probed. *Readability:* which tokens a unit stands for and how word-like its activation is. *Attribution:* whether the explanation of a prediction agrees with the effect of removing the unit. *Intervention:* how the next-token distribution changes when a unit is scaled or shifted, and how much unrelated token sets change with it. *Cost:* test perplexity and

Table 2: The last word of the prompt read through each interface (top tokens of each unit; leading hyphen: word piece).

model	interface	units that carry the last word (top tokens of each)
plain	SAE latent	[should Benny sees grandma under] ; [named came smiled was saw] ; [bear named and was .]
CoCoMix	concept	[-hood -ving -als -cing -one] ; [valuable 'd placed -red friendship] ; [does -ead pleased clos gives]
CoCoMix	SAE latent	[should under Benny grandma both] ; [came boy girl named was] ; [named bear who came -s]
Backpack	sense of *dog*	[collar leash dog bark owner] ; [-ches -gy -ner -ny -g] ; [The Rex Spot dog Max] ; [. His -ned his himself]
Backpack	SAE latent	[-Later -From -Once should upon] ; [came was is would were] ; [both -The -Mom dad -Lily]

its trajectory during training. Each probe is applied to every model that has the relevant interface; metric definitions are given in Appendix C. Section 5.1 first applies all probes to a single shared example.

5 Results

5.1 Shared Example Across Models

We use one prompt, "One day, a boy and his dog went to the park. The *dog*", and one intervention goal: to make the dog perform an action typical of dogs. All eight models predict almost the same next words (Figure 2a), so the differences below concern what can be read and done, not what is predicted. Reading the last word through each interface yields three kinds of output (Table 2). A Backpack gives four lists of tokens for *dog*: its associates (collar, leash, bark), word pieces that can follow it, the names that follow it, and function words. A CoCoMix model gives the concepts active at this position, each described by the tokens it predicts next; these are generic. A post-hoc SAE, available for any model, gives latents of the same kind as the concepts. For the intervention, we select in each interface the unit whose description shares the most tokens with the goal words ("barked wagged ran . . . ") and shift it as far as a fixed budget of collateral change allows (Figure 2b). In a Backpack the unit selected by reading is usually effective: it is one of the four senses of the word itself, and in most trained models it coincides with the best sense. For concepts and latents, the unit selected by reading is usually much less effective than the best unit in the dictionary, which is found only by exhaustive search. The following subsections analyse the causes.

5.2 Impact of Exact Decomposition on Attribution Faithfulness

Attribution. Because Eq. (1) is exact, the explanation of the prediction in Figure 3 is not an approximation: the next word *was* is carried by the previous full stop, read through one sense, and by the subject *dog*, read through another, and the contributions sum to the logit. An explanation is faithful if removing the component it names has the effect it predicts. For a Backpack this holds to rounding error: the effect of removing one (token, sense) term equals its predicted effect (Figure 4b), also when concepts are added. For post-hoc latents, whose attribution is a first-order estimate, predicted and actual effects are only loosely correlated, least so in the models with a concept channel (Figure 4a,c). A built-in explanation is exact by construction, whereas a fitted one carries no such guarantee.

Selectivity of interventions. The same exactness makes interventions selective (Figure 5, left). Scaling a sense changes the probability of that sense's own tokens smoothly with the scale, while the token sets of the word's other senses change little. This holds with the penalty and with concepts added, because the effect of a scaled sense is a known term of the sum. Sense roles are also consistent: every trained Backpack has one copy sense and one next-token sense (Appendix Figure 9), and only the index differs between models. As an application, zeroing the one sense of a profession word that carries a pronoun stereotype removes part of the he/she gap at a cost to the rest of the distribution (Appendix Figure 13). The sense differs between models and was selected on the test prompts, so this shows that the intervention exists, not that a sense index denotes "gender".

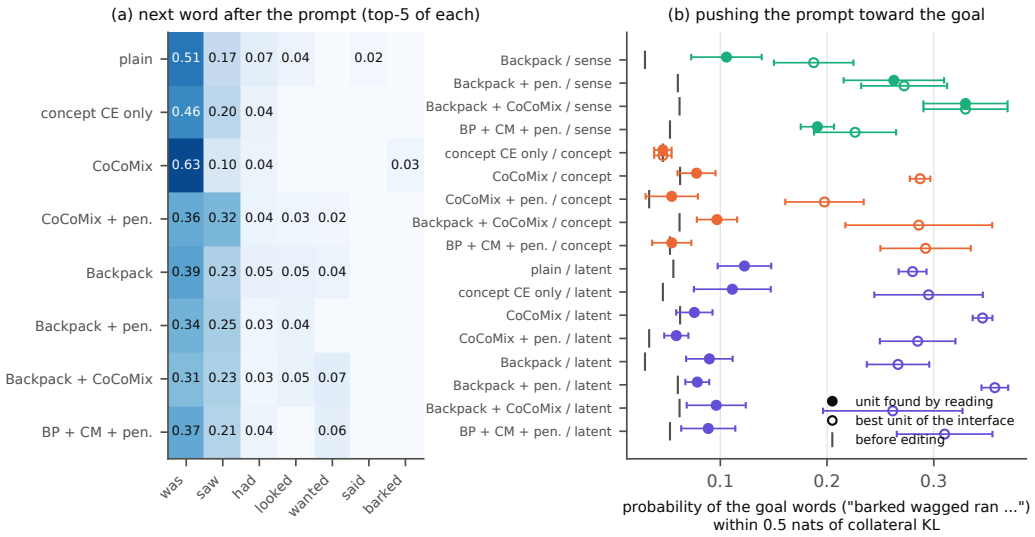


Figure 2: (a) Next-word probabilities after the prompt (top five tokens of each model). (b) Probability of the goal words after editing one unit of each interface by as much as a collateral KL of 0.5 nats allows: the unit found by reading (filled) and the best unit of the interface (open); mean and standard error over four trained models; the grey tick is the unedited value.

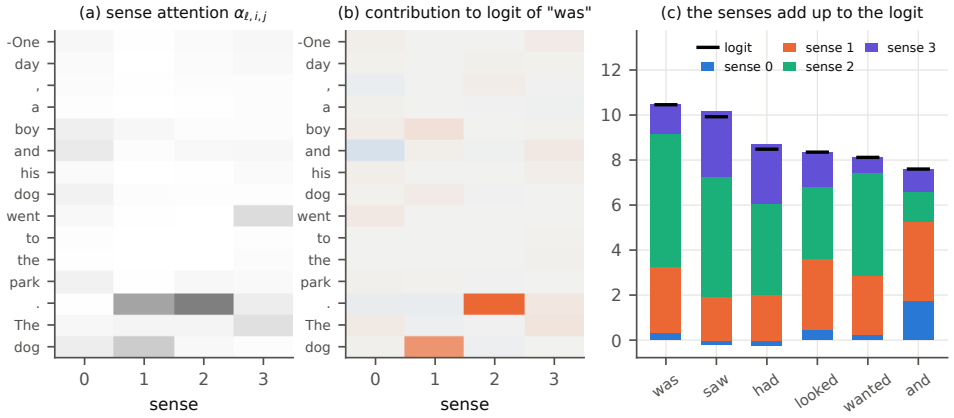


Figure 3: Exact decomposition of one prediction. (a) Attention of each sense on each source token. (b) Contribution of each (token, sense) pair to the logit of the top token. (c) Contribution of each sense to six top tokens; the stacked bars end at the logit.

5.3 Vocabulary Constraints in Concept Channels

Channel use. With a label loss only, the head learns to predict teacher concepts but the output cannot depend on it, so shifting a concept has no effect by construction (Figure 5, middle). With interleaving, the model depends on the channel completely: replacing the concepts of a story by those of another story raises the loss by several nats (Figure 6). The interleaved slots carry token information as well as concepts, which explains why the head's units are more tied to specific tokens than the teacher's (Appendix Figure 11). A channel is therefore writable only if the output is made to depend on it.

Teacher vocabulary. Channel efficacy is bounded by the teacher's vocabulary. The head predicts teacher concepts far above chance (Appendix Figure 10), but the teacher's concepts are mostly punctuation and function words, and the two most frequent are active at a quarter and a sixth of all positions. A few are readable: one fires on determiners and predicts places, another fires on copulas

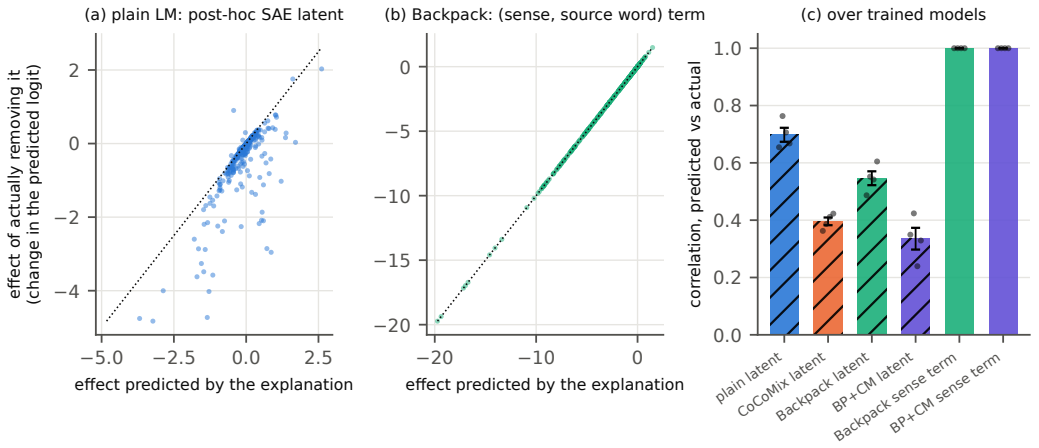


Figure 4: Does removing a unit do what the explanation says? Predicted versus actual change of the model's top logit when one unit is removed at one position. (a) Plain LM, post-hoc SAE latent, first-order estimate. (b) Backpack, one (sense, source word) term, exact. (c) Correlation over trained models.

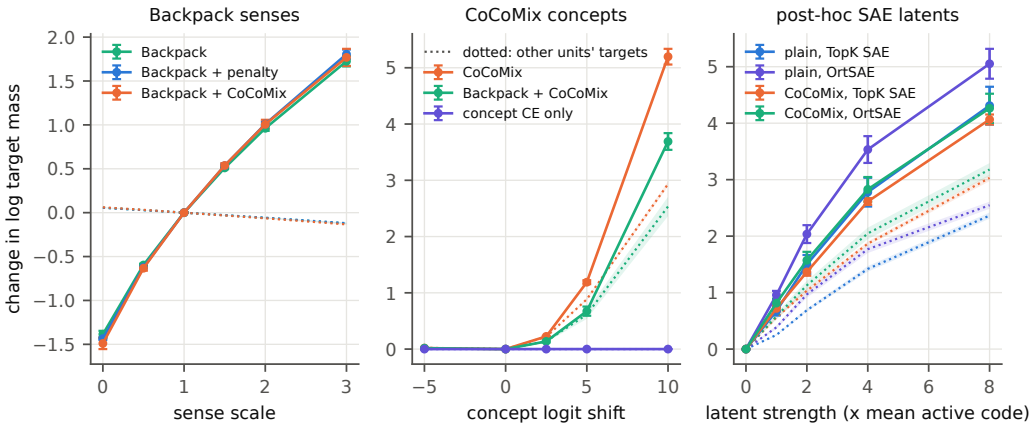


Figure 5: Change in log probability mass of a unit's own target tokens when it is intervened on (solid) and of other units' target tokens (dotted band). Left: sense scale. Middle: concept logit shift. Right: SAE latent strength. Strength axes differ; compare solid against dotted.

and predicts emotion adjectives. The concepts that matter for next-token loss concern what comes next rather than topic, so boosting one steers syntax more than meaning (Table 3): boosting the place concept moves the story outdoors but induces repetition, and boosting another breaks the text. The effect also leaks (Figure 5): raising one concept raises the token sets of unrelated concepts by about half as much, and the concept selected by reading the goal words in Figure 2 is less effective than the best one. The selectivity of a concept interface is limited by the independence of its concepts, and the teacher's concepts are entangled.

Interaction with exact decomposition. In the combined model, concepts can only re-weight senses, which preserves Eq. (1) (Figure 4c). The cost is control: the same concept shift changes the output less than in the model without senses (Figure 5). A channel that cannot rewrite the output directly is a weaker intervention.

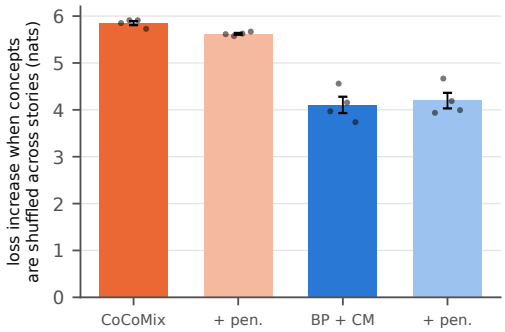


Figure 6: Increase of test loss when the concept logits of every position are replaced by those of another story. Zero for the label-only model by construction.

Table 3: Greedy continuation with +6 on one concept logit at every position; the concept is described by the tokens it predicts.

model	concept (predicts next)	continuation
CoCoMix	none (unsteered)	park with her mom. She saw a big slide and wanted to go on it. She asked her
	109: forest meadow attic	big trees. She saw a big tree with a big tree. She wanted to climb the big tree
	414: cry explore join	beach with her play in the sand. She had a big bucket and a shovel fill fill fill
	393: garden kitchen stairs	old toys inseage. She had a dolles, two dolleses, two dolles

5.4 Geometric Orthogonality versus Feature Readability

Geometric effect. The penalty achieves its objective on every kind of unit (Figure 7). SAE rows, the senses of one word and the columns that compress concepts all become much less similar to their nearest neighbour, and the penalised units fall below random vectors. The penalty designed for SAEs therefore transfers unchanged to units inside the language model. As this quantity is the penalty's own objective, however, the figure shows that the penalty acts, not that the units are better.

Behavioural effect. The behavioural measures do not follow the geometry. First-letter absorption, the failure OrtSAE targets, is nearly identical for both dictionaries (Appendix Figure 14); explained variance and the loss recovered when the SAE replaces the activations decrease slightly (Appendix Figure 12); and the word-likeness of the latents is unchanged (Appendix Figure 11). The one benefit is structural: a few TopK rows can be written as sums of OrtSAE rows, for example a place feature that splits into finer place features, whereas fewer OrtSAE rows can be written as sums of TopK rows (Appendix Table 10). Distinct entries thus yield fewer composites, but composites were not the limiting factor for readability in our setting. We hypothesise that absorption depends on how the data are partitioned among features, which a geometric penalty on the decoder does not control; we did not test this.

5.5 Computational and Routing Overheads

Table 4 and Figure 8 separate the cost of each mechanism. Adding a label loss on a head that the output does not use costs almost nothing. Feeding the head back costs little, and the gap to the plain model closes during training. Replacing the readout with senses costs more, and its gap levels off rather than closing; adding the concept channel on top costs slightly more. Wider plain models outperform every mechanism (Appendix Table 6). Quality is therefore lost where the structure changes how the output is computed, and not where it only adds an auxiliary prediction target; the effect of the penalty is below the resolution of our measurements. Compute cost follows the same pattern: every model that interleaves or uses senses trains more slowly.

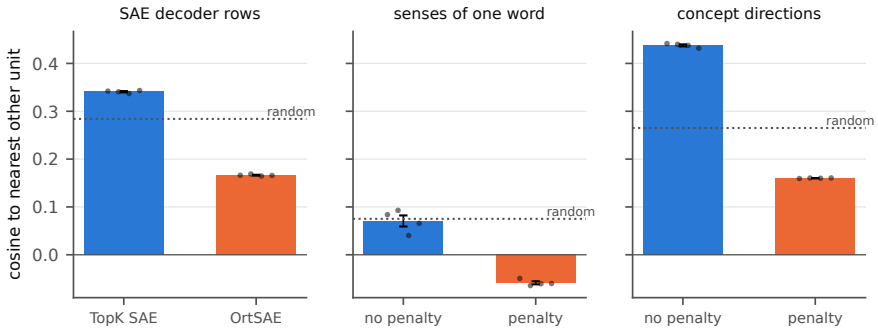


Figure 7: Cosine of each unit to its nearest neighbour without and with the penalty; dotted: random vectors.

Table 4: Parameters, training time relative to the plain model, test perplexity, paired cross-entropy difference to the plain model of the same seed, precision@4 of the concept head and steering KL; mean ± standard error over four trained models.

model	params (M)	train time	test ppl	ΔCE vs plain	concept hit@4	steering KL
plain	0.95	1.0×	7.62 ± 0.02	–	–	–
concept CE only	1.09	1.1×	7.66 ± 0.01	0.005 ± 0.003	0.39 ± 0.00	0.00 ± 0.00
CoCoMix	1.09	1.6×	7.74 ± 0.02	0.016 ± 0.002	0.37 ± 0.00	0.77 ± 0.02
CoCoMix + pen.	1.09	1.6×	7.74 ± 0.01	0.015 ± 0.003	0.37 ± 0.00	0.84 ± 0.02
Backpack	1.07	1.4×	8.04 ± 0.01	0.053 ± 0.003	–	–
Backpack + pen.	1.07	1.4×	8.06 ± 0.02	0.055 ± 0.004	–	–
Backpack + CoCoMix	1.20	1.9×	8.25 ± 0.06	0.079 ± 0.006	0.38 ± 0.00	0.35 ± 0.04
BP + CM + pen.	1.20	1.9×	8.28 ± 0.10	0.082 ± 0.013	0.38 ± 0.00	0.36 ± 0.06

6 Discussion

Source of interpretability. Of the three mechanisms, the one that yielded interpretable outputs without qualification is the one that changes the model's computation the most. The Backpack turns each prediction into an exact, attributable sum and provides selective interventions; this differs in kind from a post-hoc explanation, which can be wrong about what the model did. CoCoMix provides a readable but learned and entangled channel, which supports only coarse reading and writing. OrtSAE provides a measurable property, distinctness, that transfers to units inside the model but does not by itself make them easier to read. The shared example (Section 5.1) shows the same ordering: when the interface is a small set of exact, named parts, reading a description finds the effective intervention, whereas in a large learned dictionary the effective intervention exists but reading does not locate it. The more the structure is part of the computation, the more trustworthy the interface; the more it acts as a regulariser on a representation, the more its value depends on whether that representation was the limiting factor.

Interaction with post-hoc analysis. A post-hoc SAE explains the Backpack model best, measured by the loss recovered when its reconstruction replaces the activations, and the CoCoMix model worst; the SAE latents of both are more word-like than those of the plain model (Appendix Figures 11, 12). Built-in structure and post-hoc analysis are therefore complementary. Structure that simplifies the stream (senses carry token identity) eases the dictionary's task, whereas structure that adds a second stream (interleaved concepts) makes it harder.

Practical implications. To explain individual predictions and to edit behaviour with predictable effects, the exact additive readout is the appropriate mechanism, at a cost of a few percent of perplexity. To obtain a handle on topic-level or abstract content, a concept channel requires a better vocabulary than a same-size teacher provides, since concepts selected for next-token loss tend to concern syntax and function words. For a clean dictionary, a decoder-cosine penalty is cheap and verifiable, but should be paired with a behavioural test of the property it is meant to improve. Combining the first

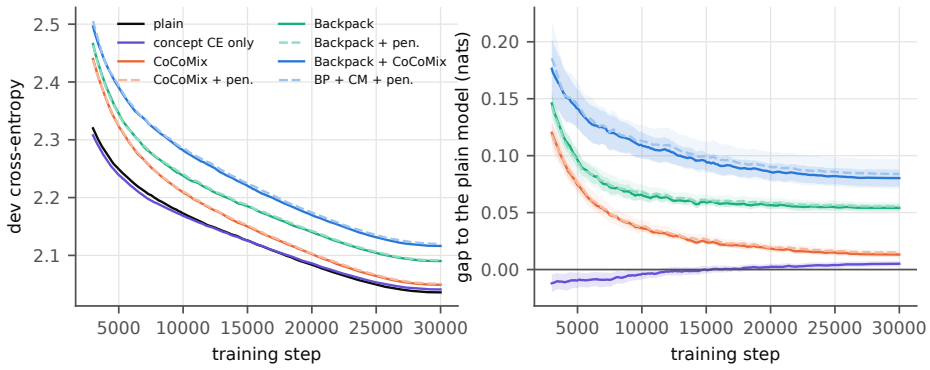


Figure 8: Left: development cross-entropy over training. Right: gap to the plain model of the same seed (band: standard error over seeds).

two mechanisms preserves exactness and weakens the concept intervention, a tension inherent to the design.

Limitations and Future Work. The study is small in every dimension: a 0.95M-parameter, two-layer model, synthetic stories, four trained models per main configuration and one for the other variants, a proxy for absorption, seven profession words for editing, and a teacher weaker than its students. Differences of a few hundredths of a nat are not claimed; the findings concern qualitative patterns (what is exact, selective or coarse, and where quality is lost). Experiments from the original papers that we did not run (topic control, knowledge editing, lexical similarity, broader SAE benchmarks) and sweeps over the number of senses, the penalty weight and the number of concepts are the natural next tests. ALl of these experiments would be feasible to do on a larger model and dataset and serve as the first step towards a future work to scale up this research project.

Broader impact. Readable and steerable models can serve bias mitigation or, equally, steering toward harmful content [Hewitt et al., 2023]. With such interpretable techniques, we can better understand model behaviour and be better at such failure modes.

7 Conclusion

Combining three built-in interpretability mechanisms in one decoder shows that the mechanisms differ in the kind of guarantee they give and in where they incur cost. The results suggest four implications for the design of interpretable architectures. First, the properties that make an interface trustworthy should be guaranteed by the architecture rather than learned: an additive readout keeps attribution and intervention exact even when other mechanisms are added, whereas a learned interface must be validated against the model it describes. Second, an interface defined over a learned vocabulary inherits that vocabulary's limits, so a concept channel needs a teacher whose concepts are selected for semantic content as well as predictive relevance, and a channel that rewrites the output competes with an exact decomposition. Third, geometric regularisers are easy to verify but verify only geometry; they should be accepted on behavioural evidence of the property they are meant to improve. Fourth, because the quality cost is localised in mechanisms that reroute computation, auxiliary supervision can be added to a model at almost no cost, and the trade-off between interpretability and quality is gradual at this scale rather than steep. More generally, our findings argue for treating interpretability mechanisms as separable components with distinct guarantees and costs, and for evaluating each against a behavioural criterion. Whether the same separation holds at larger scale, with stronger teachers and in the experiments we left out, and remains as future work.

Acknowledgements

I thank Professor Mehwish Alam for introducing me to the concepts of interpretability, for motivating me to implement the different architectures, and for supervising this project.

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A Implementation and Hyperparameters

B Deviations Between Papers and Released Code

Where a paper and its released code differ, we follow the code. CoCoMix: attribution uses post-TopK activations and the gradient of the whole-sequence mean loss with signed ranking (paper: pre-activation, per-token loss); the model reads the concept slots, which get no position embedding. OrtSAE: the Gram diagonal is zeroed (hinge at 0), the code's chunk count and optimiser differ from the text. Backpack: the released sense network has more LayerNorms and residual MLPs, warm-up is 1000 not 5000 steps, and the contextualisation formula in the paper is dimensionally inconsistent. Ours: a one-LayerNorm sense network without residual, per-token TopK instead of BatchTopK, $k{=}4$, one insertion point.

Table 5: Main settings (nothing was tuned).

LM	V=4096, d=128, 2 blocks, 2 heads, context 256, pre-LN, GELU MLP $4d$, learned positions, tied embeddings, init $\mathcal{N}(0, 0.02)$
Training	AdamW β=$(0.9, 0.95)$, weight decay 0.1, peak LR $2 \cdot 10^{-3}$, 200 warm-up steps, cosine to 0, clip 1, bf16, `torch.compile`; 30,000 steps $\times$ 128 windows $\times$ 256 tokens; four seeds for the eight main models, one for the other fifteen; checkpoint of lowest dev CE
Backpack	k=4; LayerNorm, Linear $d \to d$, GELU, Linear $d \to kd$; $W_q, W_k \in \mathbb{R}^{d \times d}$ split into k heads
CoCoMix	C=512, top-4 logits compressed, 4 labels per token, λ=0.1 (0 for interleave only)
Penalty	γ_o=0.1, one chunk, 256 random tokens per step for senses, all columns of W
SAEs	TopK, K=16, 3000 steps, batch 4096, Adam $5 \cdot 10^{-4}$, AuxK weight $1/32$, unit-norm decoder; teacher m=512 on a separately trained plain baseline; post-hoc m=1024 on the `sae` split; OrtSAE γ=0.25

C Metric definitions

- **Error.** Main models: $s/\sqrt{4}$ over the four trained models of a configuration, of a per-model statistic; other models: $s/\sqrt{n}$ over the natural samples of the measurement (test tokens, concepts, latents, letters); ratio $\sum a / \sum b = r$: $\sqrt{\sum (a_i - r b_i)^2} / \sum b_i$; effects between models use the per-token CE difference.
- **Readability.** $\max_v \sum_{t:x_t=v} a_{t,u} / \sum_t a_{t,u}$ over 1024 test windows (a: SAE code, or top-4 sparsified concept logits).
- **Geometry.** $\max_{j \neq i} \cos(w_i, w_j)$; senses: among the 4 senses of the same word; random reference: Gaussian vectors of the same number and dimension.
- **Response.** Change of $\log \sum_{v \in S_u} p(v)$ when unit u is intervened on ($+\delta$ on a concept logit at all positions; a sense scaled at all occurrences of its word; a multiple of a latent's mean active code added to h_t), averaged over affected positions. S_u: the 8 tokens with the largest $\log \frac{P(\text{next}|u)}{P(\text{next})}$ (count ≥ 15); for a sense, its 8 top vocabulary tokens. Off-target: mean over the other units' sets. 24 concepts, 24 latents, 64 sense–word pairs.
- **Common example.** Prompt "One day, a boy and his dog went to the park. The dog"; goal words: barked, wagged, bark, ran, jumped, played, licked, chased, barking, chase. A unit is described by its eight (twelve for a sense) highest-lift next tokens; the unit found by reading is the one with the largest overlap with the goal words, the best unit maximises the goal mass over the whole interface. An edit changes one unit at the last position only (sense $\times s$, concept $+s$, latent $+s\times$ its mean active code); the reported goal mass is the largest on a strength grid whose $\mathrm{KL}(p \| p')$ at that position stays ≤ 0.5 nats.
- **Faithfulness.** One random position per test window (1536 windows), the model's top token, one unit: a Backpack (sense, source word) term of the logit, or a random active post-hoc latent. Predicted: the term itself, or gradient $\times$ latent contribution; actual: the logit change when the sense is scaled to zero, or the latent's decoded contribution is subtracted from the block-0 output.
- **Concept reliance.** Test loss when the concept logits of every position are replaced by those of the same position in another window (512 windows).
- **Sense roles.** For 300 frequent words: z-score of the word in its own sense projection; Pearson correlation of the projection with $\log P(\cdot \mid w)$ (bigram) and with the positive PMI row (window ± 4), both from 20M training tokens.
- **Gender.** For 7 single-token professions in "The w said that", the share of $\sum_w |\log p(\text{he}) / p(\text{she})|$ removed by scaling one sense of w to 0; collateral KL on the full next-token distribution.
- **SAE.** Explained variance and CE recovered (block-0 output replaced by the reconstruction vs the mean) as in Korznikov et al. [2025]; first-letter probe F1 and absorption proxy (dense probe right, best single latent silent), not the SAEBench protocol; cross-dictionary decomposition by non-negative least squares (at most 3 components ≥ 0.25, cosine > 0.9).

D Full Model Comparison

Seed 0 of all 23 configurations; the eight main models are the rows of Table 4.

E Additional Results

Reproduction. I reimplemented the three methods from scratch. Code will be made available on https://github.com/shashuat/decomposability. TinyStories dataset is opensource.

Table 6: All models at 30,000 steps: parameters, test perplexity ± s.e., paired ΔCE to the plain baseline, concept precision@4, steering KL (32 concepts), and share of the he/she gap removed by the best sense (7 professions).

model	params (M)	test ppl	ΔCE vs baseline	concept hit@4	steering KL	gender gap removed
Baseline	0.95	7.614 ± 0.008	0.0000 ± 0.0000	–	–	–
Baseline d=136	1.04	7.411 ± 0.007	-0.0270 ± 0.0003	–	–	–
Baseline d=152	1.22	7.047 ± 0.007	-0.0774 ± 0.0003	–	–	–
concept CE (OrtSAE)	1.09	7.645 ± 0.008	0.0040 ± 0.0003	0.387 ± 0.000	0.00 ± 0.00	–
concept CE (TopK)	1.09	7.650 ± 0.008	0.0046 ± 0.0003	0.390 ± 0.000	0.00 ± 0.00	–
CE + interleave (OrtSAE) + ortho	1.09	7.746 ± 0.008	0.0172 ± 0.0004	0.364 ± 0.000	0.90 ± 0.16	–
CE + interleave (OrtSAE)	1.09	7.726 ± 0.008	0.0146 ± 0.0004	0.365 ± 0.000	0.80 ± 0.14	–
CE + interleave (TopK) + ortho	1.09	7.748 ± 0.008	0.0174 ± 0.0004	0.366 ± 0.000	0.81 ± 0.13	–
CE + interleave (TopK)	1.09	7.711 ± 0.008	0.0127 ± 0.0004	0.364 ± 0.000	0.83 ± 0.14	–
interleave only + ortho	1.09	7.803 ± 0.008	0.0245 ± 0.0004	–	–	–
interleave only	1.09	7.851 ± 0.008	0.0306 ± 0.0004	–	–	–
Backpack k=4 + ortho	1.07	8.026 ± 0.008	0.0527 ± 0.0004	–	–	0.56 ± 0.04
Backpack k=4	1.07	8.028 ± 0.008	0.0530 ± 0.0004	–	–	0.20 ± 0.12
Backpack k=4 + concept CE (OrtSAE) + ortho	1.20	7.936 ± 0.008	0.0414 ± 0.0004	0.388 ± 0.000	0.00 ± 0.00	0.27 ± 0.05
Backpack k=4 + concept CE (OrtSAE)	1.20	8.029 ± 0.008	0.0531 ± 0.0004	0.387 ± 0.000	0.00 ± 0.00	0.59 ± 0.06
Backpack k=4 + concept CE (TopK) + ortho	1.20	8.033 ± 0.008	0.0536 ± 0.0004	0.390 ± 0.000	0.00 ± 0.00	0.51 ± 0.10
Backpack k=4 + concept CE (TopK)	1.20	7.973 ± 0.008	0.0461 ± 0.0004	0.391 ± 0.000	0.00 ± 0.00	0.48 ± 0.07
Backpack k=4 + CE + interleave (OrtSAE) + ortho	1.20	8.379 ± 0.009	0.0957 ± 0.0004	0.376 ± 0.000	0.44 ± 0.04	0.39 ± 0.06
Backpack k=4 + CE + interleave (OrtSAE)	1.20	8.228 ± 0.009	0.0775 ± 0.0004	0.379 ± 0.000	0.35 ± 0.03	0.31 ± 0.04
Backpack k=4 + CE + interleave (TopK) + ortho	1.20	8.095 ± 0.008	0.0612 ± 0.0004	0.382 ± 0.000	0.36 ± 0.06	0.59 ± 0.06
Backpack k=4 + CE + interleave (TopK)	1.20	8.078 ± 0.008	0.0592 ± 0.0004	0.384 ± 0.000	0.30 ± 0.05	0.45 ± 0.06
Backpack k=4 + interleave only + ortho	1.20	9.046 ± 0.010	0.1723 ± 0.0004	–	–	0.39 ± 0.08
Backpack k=4 + interleave only	1.20	8.920 ± 0.009	0.1583 ± 0.0004	–	–	0.16 ± 0.05

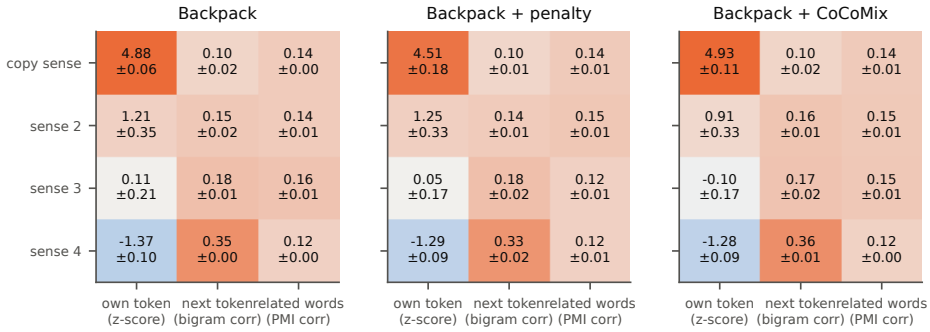


Figure 9: What each sense index does over 300 frequent words, senses ordered by how strongly they promote their own token (mean over four models): z-score of the word in its own sense projection, correlation of the projection with the bigram log-probability, and with the word's co-occurrence (PMI) row.

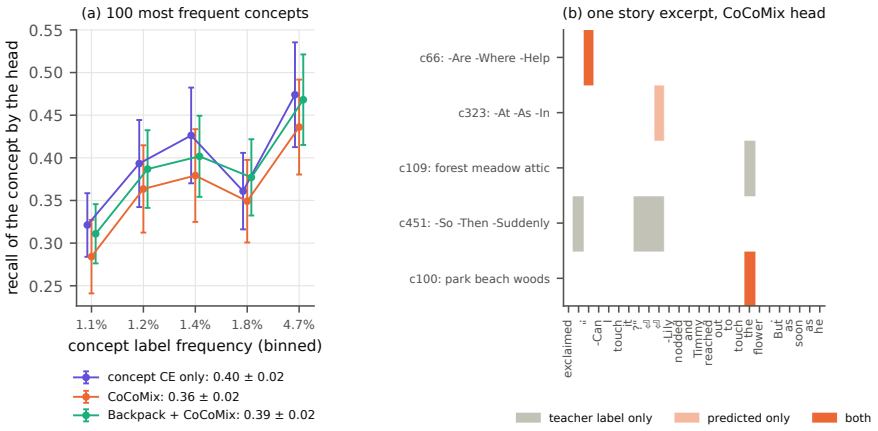


Figure 10: (a) How often the head recovers the 100 most frequent teacher concepts, by frequency bin. (b) Teacher labels and the head's predictions on one excerpt, readable concepts only.

Table 7: Top tokens of the four senses of six words, Backpack without and with the penalty (leading hyphen: word piece; seed 0).

word	sense	Backpack k=4	+ penalty (γ_o=0.1)
dog	0	collar leash dog bark owner fur	dog His . -His ! anymore
	1	-ches -gy -ner -ny -g barked	-gy -ches -g -c -ries -cy
	2	The Rex Spot dog Max Buddy	Wh Spot Rex Buddy -Spot -R
	3	. His -ned his himself to	leash owner collar fur dogs barking
happy	0	happy grateful happier glad thankful forgave	what see finding know seen that
	1	-ness -ly -ies times to -heart	-ies again to ending -by times
	2	new -iest -ier Sometimes -thday Thank	Today -old idea hard cool warm
	3	relieved proud excited -Thank . full	happy glad proud thank celebr grateful
ball	0	ball kick bounce balls kicked boun	him ! , it !" -ily
	1	-s -ing -er belonged -es -st	-sy -ing -ier -y -ed -ily
	2	it named string rope It ball	bike drum kite cand You sw
	3	." . !" and -ed !	ball boun balls bounce bounced kicked
girl	0	girl only three hair dress -Mommy	girl him She ? with her
	1	-s -av -ages 's -et named	-s -ces 's -akes -cing -ets
	2	The girl her -ail the -air	nodded The says So arrived Sometimes
	3	Her She herself her she ?	girl hair years dream three dress
play	0	play playing played turns join having	for it and thinking finding friends
	1	-ay -ill -ers -room -er nicely	-time somewhere -ened -ers -or with
	2	swings sand all toys flute rope	game flute -No piano games puzzle
	3	eating eat sing notice forget but	play played turns playing accidentally -ater
mom	0	mom 's -Mom teeth -Mommy end	-Mom -Mommy what ?" everywhere him
	1	said -age explained -ble bought replied	-hes explained gave -ning -ma reminded
	2	mom Mom kitchen -ti Her -ment	Please dad Can daddy Mommy -ache
	3	-ced daughter ? son -ened sees	sick mom early hair tummy well

Table 8: The common example for all main interfaces: unit found by reading, goal mass (mean $\pm$ s.e. over four models) for it and for the best unit, and the model's next words after the edit (seed 0).

model	lever	unit found by reading (seed 0)	goal mass	best unit	next word after
plain	SAE latent	hugged loved knew didn	0.12 $\pm$ 0.03	0.28 $\pm$ 0.01	was had saw
CoCoMix	concept	then 're enjoying hid	0.08 $\pm$ 0.02	0.29 $\pm$ 0.01	was saw barked
CoCoMix	SAE latent	replied nodded says smiled	0.08 $\pm$ 0.02	0.35 $\pm$ 0.01	was saw barked
Backpack	sense of *dog*	collar leash dog bark	0.11 $\pm$ 0.03	0.19 $\pm$ 0.04	was saw barked
Backpack	SAE latent	daddy say gave ran	0.09 $\pm$ 0.02	0.27 $\pm$ 0.03	was saw and
Backpack + CoCoMix	sense of *dog*	dog bark leash collar	0.33 $\pm$ 0.04	0.33 $\pm$ 0.04	was saw barked
Backpack + CoCoMix	concept	then 're enjoying hid	0.10 $\pm$ 0.02	0.29 $\pm$ 0.07	was saw wanted
Backpack + CoCoMix	SAE latent	go -They ran dog	0.10 $\pm$ 0.03	0.26 $\pm$ 0.07	saw was came

Table 9: Teacher concepts: tokens where the concept fires, tokens it predicts next, recall by the CoCoMix head. Top: two most frequent; bottom: seven with the sharpest predicted tokens.

concept	positions (%)	fires on	predicts next	recall (CoCoMix)
168	22.7	int reg frown offered	End -elly middle end	0.90 $\pm$ 0.001
462	16.8	mic unpack -ret wake	valuable 'd placed -red	0.85 $\pm$ 0.001
66	0.9	?? ?? " '	-Are -Where -Help -Can	0.27 $\pm$ 0.005
323	1.6	↩ min -F rubb	-At -As -In -Their	0.59 $\pm$ 0.004
109	0.8	especially the its this	forest meadow attic garden	0.29 $\pm$ 0.005
451	0.9	? won ! -	-So -Then -Suddenly -Finally	0.10 $\pm$ 0.003
162	0.8	were 're became are	pleased welcome delighted disappointed	0.29 $\pm$ 0.005
274	1.5	↩ late -ha ex	-Her -His -Everyone -Their	0.74 $\pm$ 0.004
100	1.3	the The -ig dra	park beach woods sunshine	0.62 $\pm$ 0.004

Table 10: TopK decoder rows written as sums of OrtSAE rows (plain LM), each described by its top tokens.

TopK latent (top tokens)	OrtSAE components (coefficient × top tokens)	cosine
[". !". .]	0.84×[". !". ?".] + 0.43×[. ." ".]	0.94
[a A -A]	0.86×[a A -A] + 0.30×[-A A a]	0.92
[couch street ground]	0.60×[jungle yard garden] + 0.48×[war mountain ground] + 0.30×[front -ha ear]	0.94

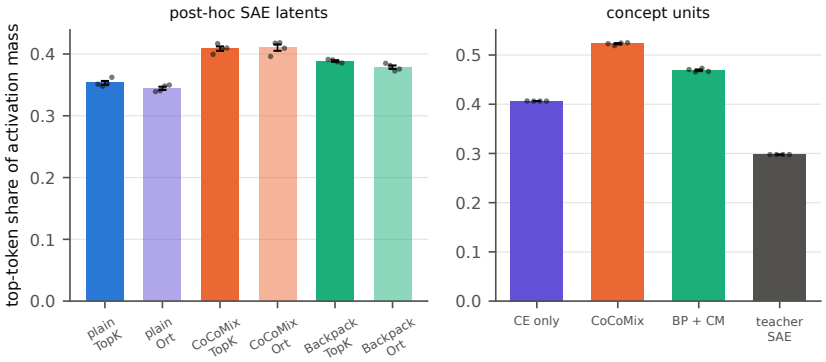


Figure 11: How word-like are the units: share of a unit's activation on its top token, averaged over units (bars: mean over four models; dots: single models). Left: post-hoc SAE latents. Right: concept units and the teacher's latents.

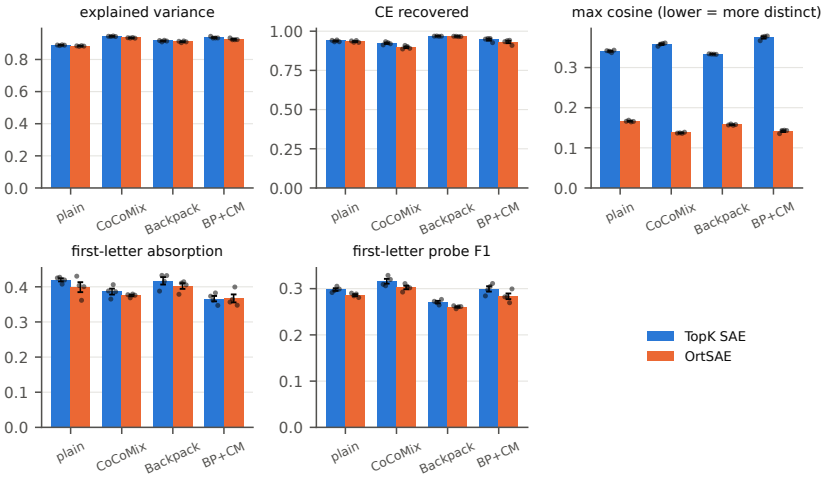


Figure 12: Post-hoc SAE metrics, TopK against OrtSAE (mean ± s.e. over four models; dots: single models). BP+CM: Backpack + CoCoMix.

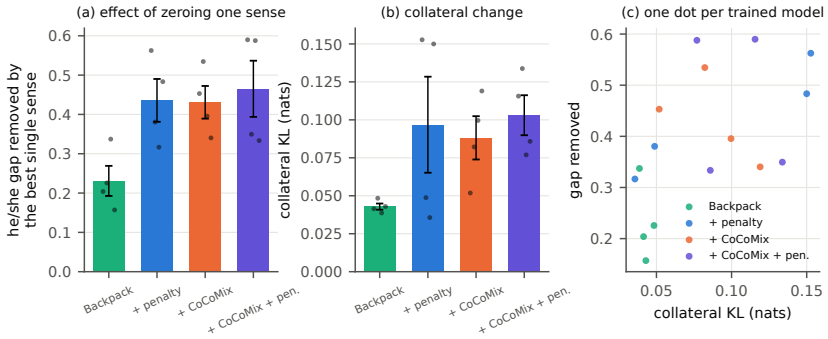


Figure 13: Zeroing one sense of seven profession words, best sense of each model (chosen on the same prompts, so an upper bound). (a) Fraction of the he/she gap removed. (b) Collateral KL on the next-token distribution. (c) One dot per trained model.

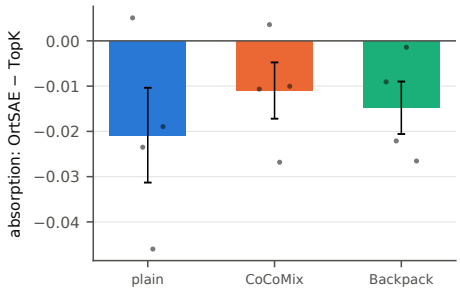


Figure 14: First-letter absorption, OrtSAE minus TopK (mean $\pm$ s.e. over four models; dots: single models).

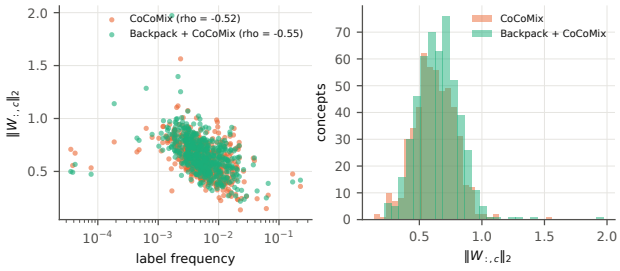


Figure 15: Compression-matrix column norms against label frequency of the concept (Spearman -0.52 CoCoMix, -0.55 Backpack + CoCoMix).

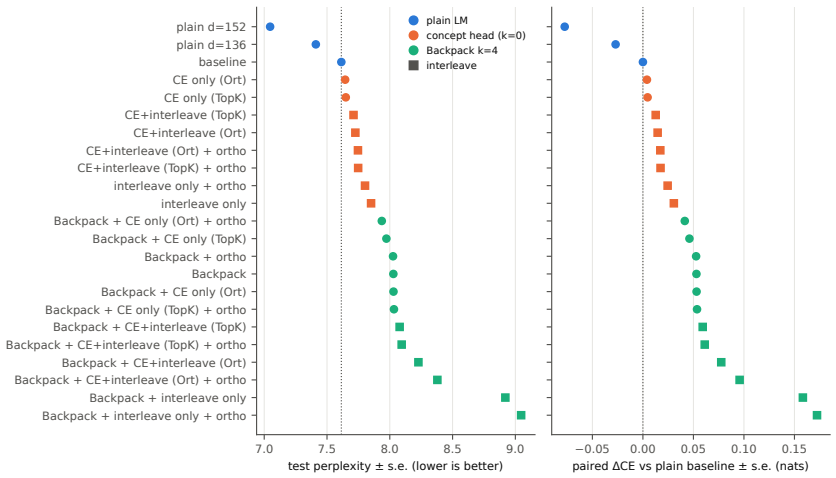


Figure 16: Test perplexity and paired ΔCE of all 23 models ($\pm$ s.e.; bars are smaller than the markers).

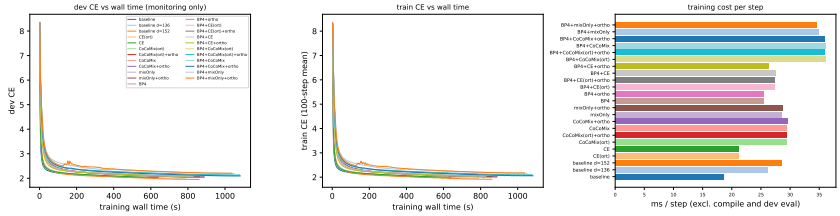


Figure 17: Cost of the training runs of the 23 seed-0 models: dev CE and train CE against wall-clock time, and milliseconds per step (18.5–36.1 ms; one RTX A5000).